

Toward Efficient Consistency Models via Variance-Reduced Distillation

Useok Choi^{ID}, Seungjoon Lee^{ID}, and MyeongAh Cho^{ID}

Abstract—Diffusion models have demonstrated remarkable performance across a wide range of generative tasks; however, their high sampling cost remains a critical bottleneck. To address this, consistency distillation (CD) was proposed, offering a reduction in sampling cost by distilling a pretrained diffusion model. However, achieving generative quality comparable to diffusion models requires extensive training for the distillation process, posing a substantial computational challenge. In this article, we introduce variance-reduced consistency learning (vrCL), a novel distillation technique that enables stable and efficient training of consistency models without relying on teacher model evaluations. By leveraging a student-guided sample pair, vrCL ensures training stability while significantly reducing computational costs. This design eliminates the need for repeated teacher model evaluations during training, resulting in high computational efficiency and significantly reduced training time. Empirical results demonstrate that vrCL achieves competitive generative performance with high training efficiency, reaching strong results within just 100k training iterations.

Index Terms—Accelerated training, diffusion models, generative models, knowledge distillation, variance reduction.

I. INTRODUCTION

DIFFUSION models [1], [2], [3] have emerged as a powerful class of generative models, leveraging probabilistic flow (PF) ordinary differential equations (ODEs) to model the data distribution through score-based learning [4]. By employing numerical ODE solvers, these models iteratively refine noise into high-quality samples. Despite their impressive performance, diffusion models inherently suffer from high computational costs and error accumulation due to iterative numerical integration. To mitigate these issues, recent advancements in numerical solvers [5], [6], [7], [8], [9] have reduced the sampling steps to around 10, significantly improving efficiency. These improvements have expanded their practical applications in high-fidelity image synthesis [10], [11], [12], [13], [14], [15], but the reliance on iterative refinement remains a bottleneck in real-time generative tasks.

To alleviate these issues, recent studies have proposed distillation techniques to enable efficient one-step sampling,

Received 18 August 2025; revised 6 January 2026 and 31 March 2026; accepted 15 June 2026. This work was supported in part by the National Research Foundation of Korea (NRF) funded by Korean Government (MSIT) under Grant RS-2024-00456589 and in part by Korea Planning and Evaluation Institute of Industrial Technology (KEIT) funded by Korean Government (MOTIE) (Development of AI Technology Capable of Robustly Recognizing Abnormal and Dangerous Situations and Behaviors During Night and Bad Weather Conditions) under Grant RS-2024-00442120. (Corresponding author: MyeongAh Cho.)

The authors are with the Department of Software Convergence, Kyung Hee University, Republic of Korea (e-mail: maycho@khu.ac.kr).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TNNLS.2026.3705500>, provided by the authors.

Digital Object Identifier 10.1109/TNNLS.2026.3705500

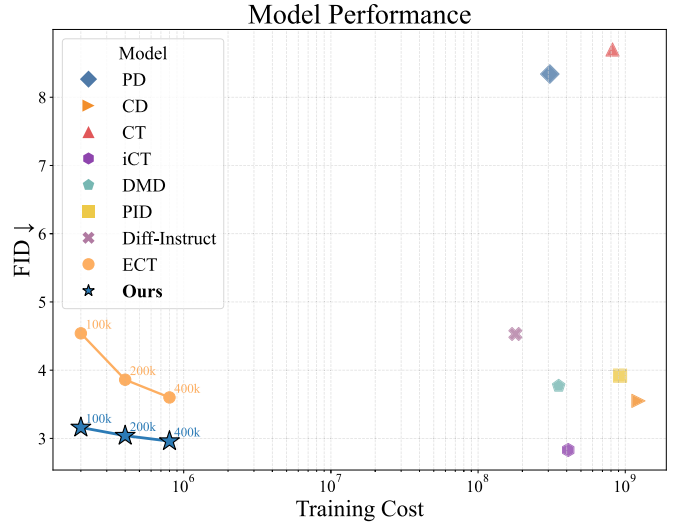


Fig. 1. Comparison of state-of-the-art generative models. Training cost is measured as the product of training images and network evaluation per iteration.

eliminating the need for iterative integration. Variance score distillation (VSD) techniques [16], [17], [18], [19], [20], [21], [22], [23] leverage pretrained diffusion models to supervise a one-step generator. These methods typically generate fake samples from a one-step generator G_θ and train a separate fake score network to match the scores of a pretrained diffusion model. Through joint training, the generator G_θ is updated based on feedback from the score matching objective, progressively improving its generative capability. Despite efficiency gains at inference, VSD methods heavily rely on auxiliary networks and often incorporate additional loss terms—such as adversarial losses—which add significant complexity and further dependencies to the training process.

In contrast, consistency models [24], [25], [26], [27], [28] have been proposed as an alternative paradigm that enforces self-consistency along the PF ODE trajectory. They train a consistency function that allows direct one-step generation. There are two primary training strategies for consistency models: consistency distillation (CD) and consistency training (CT). CD follows a diffusion-supervised approach, utilizes a pretrained diffusion model to approximate the true score function $\nabla \log p(\mathbf{x}_t)$ of data distribution, ensuring training stability by relying on low-variance and accurate score estimation. However, CD requires evaluation from a pretrained diffusion model at every iteration during training, resulting in significant computational overhead and increased training cost.

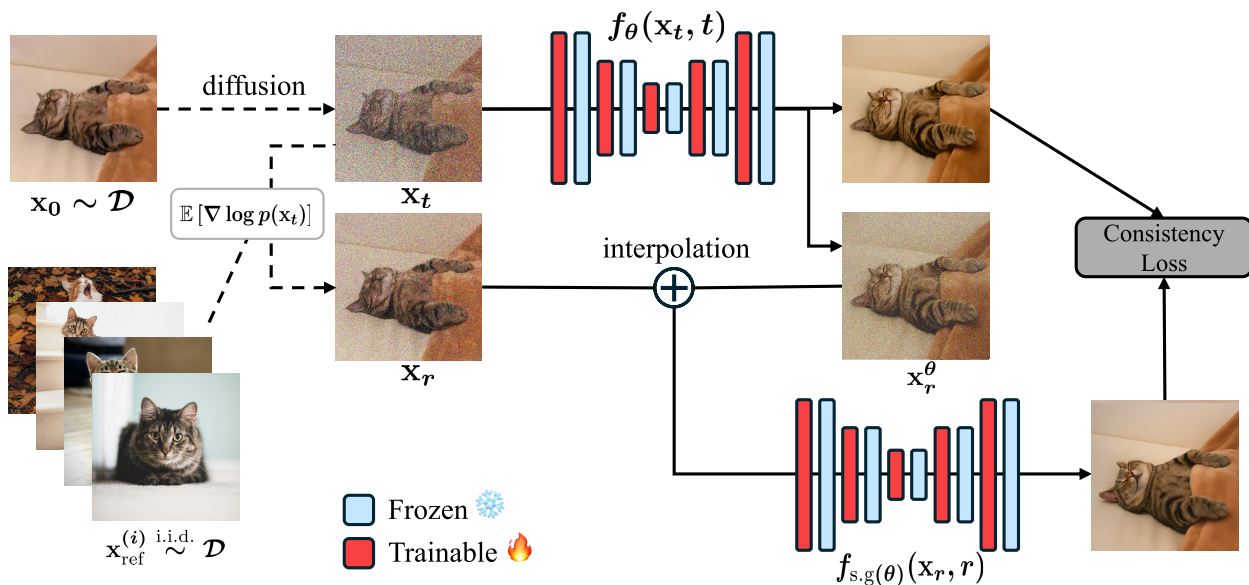


Fig. 2. Overview of our vrCL framework. We generate a noisy sample $\mathbf{x}_t$ via a forward diffusion process. To construct the $\hat{\mathbf{x}}_r^\theta$, we interpolate between a reference-based stable estimation of $\mathbf{x}_r$, estimated by expected score $\mathbb{E}[\nabla \log p(\mathbf{x}_r)]$, and the student-guided sample $\mathbf{x}_r^\theta$. Reference samples $\mathbf{x}_{\text{ref}}^{(i)}$ are independently drawn from the data distribution $\mathcal{D}$ and are used to estimate the expected score. Red and blue blocks denote trainable and frozen network parameters, respectively.

On the other hand, CT directly optimizes the consistency model using an unbiased score estimator, without relying on approximated scores from a pretrained diffusion model, making it more flexible and computationally efficient. However, CT suffers from high-variance score estimation, leading to unstable training in the early stages and slower convergence compared to CD. These limitations make it challenging for CT to achieve competitive performance without extensive training. Nevertheless, since CT does not rely on a pretrained diffusion model, it avoids potential inaccuracies or biases inherent in the teacher’s score estimation. This allows CT to achieve a higher performance upper bound by directly learning the consistency function, as also observed in recent works [25], [29], [30], [31].

In this article, we propose a novel distillation method for training consistency models. Our approach formulates a unified framework that bridges CD and CT through a continuous interpolation mechanism. Rather than a direct combination, it reinterprets the relationship between CD and CT within a single training formulation. Our approach utilizes *two different estimations* of $\mathbf{x}_r$ and dynamically interpolates between them during training. The first, $\mathbf{x}_r^\theta$, is a *student-guided reference sample* analogous to that used in CD. Instead of relying on an external teacher model for score supervision like CD, we construct $\mathbf{x}_r^\theta$ directly using the student’s own predictions. This eliminates the overhead of repeated forward passes through a pretrained diffusion model by *reusing* the student’s own predictions $f_\theta(\mathbf{x}_t, t)$ via an implicit ODE traversal, offering greater stability in the early stages of training compared to CT.

As training progresses, we gradually increase reliance on unbiased score estimation $\mathbf{x}_r$. This training strategy inherits the benefits of CT, such as the potential for higher performance than CD, while mitigating the bias that may arise from student-guided reference sample $\mathbf{x}_r^\theta$ in the later stages of training. Through this adaptive strategy, our approach effectively combines the strengths of both CT and CD, enabling efficient

and accelerated distillation. Furthermore, we introduce two specialized techniques that enhance score estimation stability and accelerate the distillation process. By integrating these techniques, our framework achieves a computationally efficient training pipeline for high-quality generative modeling.

We empirically show that our method, termed variance-reduced consistency learning (vrCL), achieves competitive generative performance with significantly reduced training cost requirements on CIFAR-10 and ImageNet. As shown in Fig. 1, vrCL achieves strong performance while maintaining low training costs and high learning efficiency. Notably, it attains a two-step FID of 2.02 on CIFAR-10 within just 100k training iterations. The overall framework of vrCL is illustrated in Fig. 2.

II. RELATED WORKS

Diffusion models [1], [32] have emerged as a dominant paradigm in generative modeling, learning to reverse a forward noising process by estimating the score function of the data distribution [4]. Their success has rapidly expanded across a wide range of domains, including high-fidelity image synthesis [33], [34], [35], image editing and stylization [11], and even scientific domains such as 3-D molecular generation [10]. Despite their high generation quality, they suffer from slow sampling speed, typically requiring tens to thousands of iterative denoising steps.

Various acceleration techniques, such as DDIM [2], higher-order ODE solvers [6], [36], and progressive distillation technique [17], aim to reduce the sampling steps. However, these methods often exacerbate trajectory mismatch between the training SDE/ODE and the accelerated inference path, resulting in noticeable degradation in perceptual quality.

Consistency models offer an alternative to diffusion models by learning to map noisy samples on the probabilistic flow ordinary differential equation (PF-ODE) trajectory directly to their denoised endpoints. The foundational training paradigms

include CD, where a pretrained diffusion model provides teacher guidance, and CT, which learns from scratch without teacher supervision. Early studies CM [24] reported that CT underperforms CD, primarily due to instability when trained without external guidance.

However, recent works [25], [29], [31] have shown that CT can achieve competitive or even superior performance compared to CD. For example, iCT [25] enables student-only training to outperform CD in certain settings, although often at higher computational cost. To mitigate this, ECT [30] reinterprets the denoising score matching (DSM) loss as a generalized form of the consistency objective, leading to a two-phase training framework. While this improves both performance and efficiency, it does not fully resolve the issue of unstable score estimation in CT.

Another line of research focuses on extending consistency models to multistep generation. CTM [27] introduces a DDIM-inspired parameterization by embedding both the source timestep t and the target timestep s , while PCM [28] segments the PF-ODE trajectory into discrete intervals and performs CD within each segment. Despite these advancements, such methods often involve more complex training procedures and additional components, which can undermine the simplicity and efficiency of consistency models.

III. PRELIMINARIES

A. Diffusion Models

Diffusion models are a class of probabilistic generative frameworks that model the continuous-time transformation between the data distribution p_{data} and a predefined prior distribution (commonly an isotropic Gaussian). This transformation can be expressed as a stochastic process in forward time, which gradually perturbs the data with Gaussian noise until it becomes indistinguishable from the prior. The reverse of this process corresponds to a denoising generative procedure.

The deterministic counterpart of this stochastic process is described by the probability flow ordinary differential equation (PF-ODE), which represents the marginal probability evolution without stochasticity

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2}g^2(t)\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) \quad (1)$$

where $\mathbf{f}(\mathbf{x}_t, t)$ denotes the drift term determined by the forward process, $g(t)$ is the diffusion coefficient controlling noise injection, and $\nabla \log p(\mathbf{x}_t)$ is the *score function*—the gradient of the log-density of the noisy data distribution at time t .

In practice, a neural network parameterized by θ is trained to approximate this score function. Sample generation is then performed by numerically integrating the PF-ODE backward in time from $t = 1$ (pure noise) to $t = 0$ (clean data). While diffusion models excel in generation quality, they require tens to thousands of integration steps, making them computationally expensive at inference due to their inherently sequential denoising nature.

B. Consistency Models

Consistency models are proposed to overcome the slow sampling bottleneck of diffusion models by bypassing the need

for incremental integration. Instead of simulating the PF-ODE step-by-step, a consistency model learns a mapping

$$f_{\theta}(\mathbf{x}_t, t) \rightarrow \mathbf{x}_0$$

that predicts a consistent endpoint (e.g., the clean sample $\mathbf{x}_0$) directly from any noisy state $\mathbf{x}_t$ along the PF-ODE trajectory. This property—known as the *consistency property*—requires that predictions made from different points along the same trajectory agree with each other.

Formally, the consistency matching loss is expressed as follows:

$$\mathcal{L}^{\text{CM}}(\theta) = \mathbb{E}[d(f_{\theta}(\mathbf{x}_t, t), f_{sg(\theta)}(\mathbf{x}_r, r))] \quad (2)$$

where $\mathbf{x}_t$ and $\mathbf{x}_r$ ($r < t$) are two states sampled from the same PF-ODE trajectory, $d(\cdot, \cdot)$ denotes a distance metric measuring the discrepancy between two model predictions, and $sg(\theta)$ is the *stop-gradient* operator to prevent trivial solutions. In existing consistency-based frameworks, different forms of $d(\cdot, \cdot)$ have been adopted, including combinations of ℓ_2 and LPIPS losses in early consistency models [24], the pseudo-Huber loss in iCT [25], and adaptive-weighted ℓ_2 objectives in recent approaches such as ECT [30].

There are two primary training paradigms for consistency models, distinguished by whether the student receives guidance from a pretrained diffusion model.

1) *CD*: In CD, $\mathbf{x}_t$ is obtained by perturbing a clean sample $\mathbf{x}_0 \sim \mathcal{D}$ with the forward perturbation kernel $q(\mathbf{x}_t|\mathbf{x}_0)$, and $\mathbf{x}_r$ is computed by applying a reverse PF-ODE step using an ODE solver guided by a pretrained diffusion model. This provides low-variance score information and stable target pairs $(\mathbf{x}_t, \mathbf{x}_r)$, which can lead to a more stable training process and faster convergence, but it requires repeated teacher evaluations during each training iteration, resulting in high computational overhead and limited scalability.

2) *CT*: CT eliminates the teacher model entirely. Here, $\mathbf{x}_r$ is estimated using an unbiased score approximation, e.g.,

$$\mathbf{x}_r \approx \mathbf{x}_0 + r \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

which corresponds to the reverse PF-ODE step guided only by the analytic score estimation. Unlike CD, which computes $\mathbf{x}_r$ by integrating the reverse PF-ODE under a pretrained diffusion teacher at every iteration, CT approximates $\mathbf{x}_r$ without any teacher evaluations. Consequently, CT is computationally efficient and not constrained by the teacher's quality ceiling. However, the unbiased score estimator exhibits high variance, which in turn causes optimization instability and slower convergence relative to CD.

In summary, CD offers stability via external guidance but at the cost of efficiency, whereas CT offers efficiency and potential performance gains but at the cost of stability and requiring more training iterations. This tradeoff has motivated numerous recent works to seek hybrid or enhanced training strategies that inherit the strengths of both paradigms.

To better understand the practical implications of using CD versus CT, we provide a detailed quantitative analysis of their training dynamics. While CD benefits from stable supervision, it incurs significant computational cost due to teacher evaluations. In contrast, CT is computationally efficient but may suffer from instability in early training.

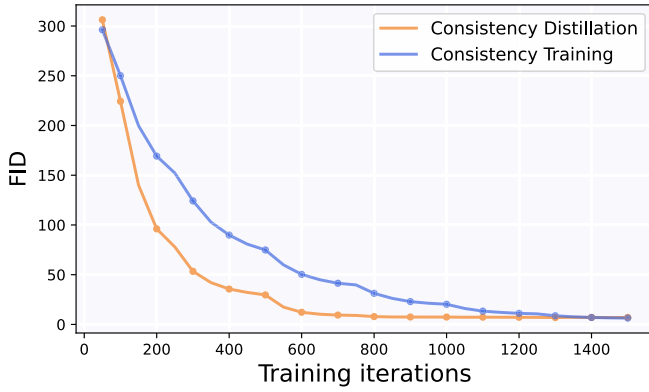


Fig. 3. FID comparison between CD and CT during training on CIFAR-10.

TABLE I
TRAINING STATISTICS FOR CD AND CT ON CIFAR-10

Method	sec/tick	sec/kimg	Iterations	GFLOPs	FID
CD	66.3	5.43	80,000	5.11M	7.89
CT	52.1	4.09	135,000	5.75M	7.78

For a fair comparison between CD and CT, we initialize both models with the same pretrained diffusion model and apply identical configurations, including timestep scheduling, noise distribution, and other training hyperparameters. We log the one-step FID score alongside key statistics such as seconds per tick, seconds per kimg, total iterations, and GFLOPs required to reach competitive performance in terms of FID score.

As shown in Fig. 3, CD achieves faster convergence and more stable training compared to CT, reaching a competitive FID within 80k iterations—equivalent in performance to CT at around 135k iterations. However, despite requiring fewer iterations, CD incurs a similar computational cost due to the overhead of evaluating the pretrained diffusion model. As summarized in Table I, this is also reflected in the training time per kimg: CD takes 5.43 s, while CT takes only 4.09 s.

Notably, the computational gap between CD and CT becomes more pronounced as model size, dataset scale, or training iteration count increases. Since CD requires repeated evaluations of a large pretrained diffusion model at every step, the overhead scales significantly with training cost. This bottleneck is particularly critical in modern generative modeling pipelines, which employ high-capacity architectures trained on large-scale datasets over hundreds of thousands of iterations. Moreover, recent CD-based consistency models often perform multiple evaluations per step to ensure accurate score estimation, further exacerbating the computational burden. As a result, while CD offers early-stage stability, its high computational demand can limit scalability and practicality in real-world applications. These observations highlight the importance of designing distillation frameworks that retain stability without incurring excessive cost—one of the key motivations behind our proposed method.

IV. VARIANCE-REDUCED CONSISTENCY LEARNING

Existing training methods for consistency models often incur high distillation costs. CD provides stable supervision via pretrained diffusion models but is computationally expensive. In contrast, CT is more efficient but often suffers from instability in the early stages of training. These limitations present a significant bottleneck in the effective training of consistency models. To address this, we propose a simple yet effective training method that combines the efficiency of CT with the stability of CD, thereby bridging the strengths of both approaches.

A. Student-Guided CD

CD supervises score estimation using reverse-time samples $\mathbf{x}_r$, which are obtained by numerically solving the PF-ODE from $\mathbf{x}_t$ under the guidance of the teacher’s score function $s_\phi(\mathbf{x}_t, t) \approx \nabla \log p(\mathbf{x}_t)$. In modern generative modeling pipelines, where high-capacity networks are trained over millions of iterations on large-scale image datasets, the per-step evaluation of a pretrained teacher model poses a major computational burden. To address these limitations and improve the efficiency of consistency model training, we modify the conventional distillation process by removing the reliance on teacher evaluation. Instead, we construct a low-variance reverse-time estimate from $\mathbf{x}_t$ using the student model via an Euler- or DDIM-like parameterization of the PF-ODE.

$$\begin{aligned}
 \mathbf{x}_r^\theta &= \mathbf{x}_t + \int_t^r \frac{\mathbf{x}_u - \mathbb{E}[\mathbf{x}_0|\mathbf{x}_u]}{u} du \\
 &\approx \mathbf{x}_t + \int_t^r \frac{\mathbf{x}_t - f_\theta(\mathbf{x}_t, t)}{t} du \\
 &\approx \mathbf{x}_t + \frac{\mathbf{x}_t - f_\theta(\mathbf{x}_t, t)}{t} \int_t^r du \\
 &\approx \frac{r}{t} \mathbf{x}_t + \left(1 - \frac{r}{t}\right) f_\theta(\mathbf{x}_t, t). \tag{3}
 \end{aligned}$$

A key advantage of this formulation is that the denoised prediction, used for the consistency matching loss, is also reused to generate the reverse-time sample, eliminating the need for costly network evaluations. This placement directly links Fig. 4 to the reverse-time reference-sample generation process depicted in the figure and places its first citation after Fig. 3 and before Figs. 5–7. While CD requires at least three forward passes per step—one for the teacher score evaluation $s_\phi(\mathbf{x}_t, t)$, and two for the student $f_\theta(\mathbf{x}_t, t)$ and $f_{sg(\theta)}(\mathbf{x}_r, r)$ —recent generative methods often involve multiple teacher evaluations for score refinement, further increasing the computational cost. In contrast, our method eliminates the need for teacher supervision entirely, requiring only two student evaluations. This matches the computational cost of CT while providing more stable sample pair generation by avoiding high-variance score estimates.

B. Learning With Dual Reference Samples

While our parameterized generation of $\mathbf{x}_r^\theta = r/t\mathbf{x}_t + (1 - r/t)f_\theta(\mathbf{x}_t, t)$ provides a teacher-free mechanism to traverse the PF-ODE trajectory, it may introduce approximation error, as it does not explicitly approximate the score field like diffusion models—which learn the continuous vector field of

the PF-ODE—whereas consistency models only learn endpoint mappings. In early training, this formulation remains reliable due to the student being initialized from a pretrained diffusion model. As training progresses, $\mathbf{x}_r^\theta$ may slightly deviate from the ideal trajectory. To mitigate this, we adopt an adaptive strategy that interpolates between two different $\mathbf{x}_r$ estimates: student-guided reference sample $\mathbf{x}_r^\theta$, and unbiased score estimate sample $\mathbf{x}_r$, given by $-(\mathbf{x}_t - \mathbf{x}_0)/t^2$ following CT. While the latter provides a statistically unbiased score supervision signal, it is known to suffer from high variance, which can destabilize training. In contrast, the former offers greater stability, particularly in the early training phase. We define the final training sample $\hat{\mathbf{x}}_r^\theta$ as a convex combination

$$\hat{\mathbf{x}}_r^\theta = \beta \mathbf{x}_r + (1 - \beta) \mathbf{x}_r^\theta. \quad (4)$$

In the early stages of training, $\mathbf{x}_r^\theta$ tends to be more accurate because it leverages the pretrained diffusion model’s well-established local denoising capabilities, providing smoother and more stable supervision. As training progresses, however, the unbiased estimate $\mathbf{x}_r$ becomes increasingly precise, capturing the correct global denoising direction without being limited by the student’s initial bias. To maintain training stability throughout, we therefore begin with a stronger reliance on $\mathbf{x}_r^\theta$ and gradually shift toward $\mathbf{x}_r$, allowing the model to benefit from both stable early-phase guidance and high-fidelity late-phase refinement.

To maintain training stability throughout, we employ a dynamic scheduling strategy that gradually increases reliance on the unbiased estimate sample $\mathbf{x}_r$ as training progresses. Specifically, we define the effective training reference sample $\hat{\mathbf{x}}_r^\theta$ as a weighted sum between two samples $\mathbf{x}_r, \mathbf{x}_r^\theta$, with the weight adjusted over time by a scheduling function. Inspired by prior works on timestep scheduling in consistency models [25], [30], which have shown that piecewise or staircase-style schedules can stabilize learning dynamics, we adopt a staircase scheduling function, where k is the current training iteration, K is the total number of training iterations, and s is a hyperparameter that controls the growth rate of the schedule. The step interval is defined as $I_K = K/n$, where n denotes the number of growth steps. This scheduling determines how quickly the interpolation shifts from the student-generated reference $\mathbf{x}_r^\theta$ to the unbiased estimate $\mathbf{x}_r$.

$$\beta(k) = \tanh \left(s \cdot \left\lfloor \frac{k}{I_K} \right\rfloor \right), \quad I_K = \frac{K}{n}. \quad (5)$$

Although both estimates approximate the same PF-ODE reverse sample, they offer complementary supervision signals—one derived from model predictions, the other from analytical approximation. This hybrid scheme effectively unifies the benefits of CD and CT: it preserves the stable early-phase dynamics of CD, while ultimately enabling the model to surpass the performance limitations of teacher-driven supervision through the flexible and data-driven generalization of CT. As illustrated in Fig. 7, our method exhibits significantly improved stability in score estimation across training, especially in early stages. We also provide the full algorithmic description of our training procedure in Algorithm 1 to facilitate reproducibility and clarity.

Algorithm 1 vrCL

Input: Dataset $\mathcal{D}$, pretrained diffusion model ϕ , growth rate $\beta(\text{Iters})$, weighting function $w(t)$

Initialize: $\theta \leftarrow \phi$, Iters $\leftarrow 0$

repeat

 Sample $\mathbf{x}_0 \sim \mathcal{D}$, $\epsilon \sim p(\epsilon)$, $t \sim p(t)$, $r \sim p(r)$

 Compute $\mathbf{x}_t = \mathbf{x}_0 + t \cdot \epsilon$, $\mathbf{x}_r = \mathbf{x}_0 + r \cdot \epsilon$

 Compute $\mathbf{x}_r^\theta = \frac{r}{t} \mathbf{x}_t + (1 - \frac{r}{t}) f_\theta(\mathbf{x}_t, t)$

 Compute $\hat{\mathbf{x}}_r^\theta = \beta \mathbf{x}_r + (1 - \beta) \mathbf{x}_r^\theta$

 Compute loss: $L(\theta) = w(t) \cdot d(f_\theta(\mathbf{x}_t), f_{\text{sg}(\theta)}(\hat{\mathbf{x}}_r^\theta))$

 Update: $\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$

 Increment: Iters $\leftarrow$ Iters + 1

until convergence

return θ

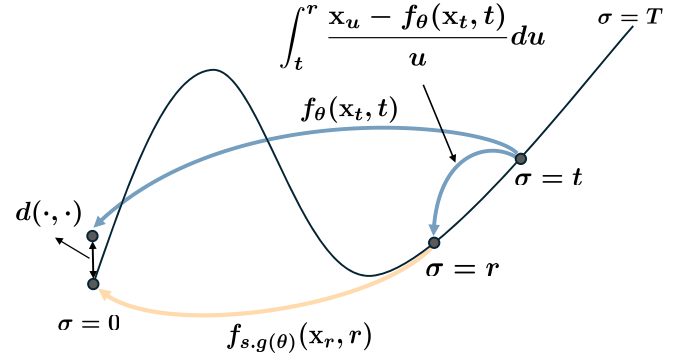


Fig. 4. Illustration of reverse-time reference sample $\mathbf{x}_r^\theta$ generation. Our student-guided parameterization enables $\mathbf{x}_t \rightarrow \mathbf{x}_r$ estimation without teacher supervision, offering stable supervision along the PF-ODE trajectory.

V. EFFECTIVE STRATEGIES FOR VRCL

To further enhance the compatibility and performance of our proposed distillation framework, we introduce two auxiliary techniques specifically designed to complement our method. The first technique, partial parameter freezing, facilitates rapid distillation by reducing the number of trainable parameters, thereby accelerating convergence and lowering training cost. The second technique addresses the instability that may arise from the use of an unbiased score estimator. We propose a stabilization method that improves the reliability of the unbiased score estimation $-(\mathbf{x}_t - \mathbf{x}_0)/t^2$ during training. Together, these techniques improve both the stability and efficiency of our distillation process.

A. Partial Parameter Freezing for Accelerated Distillation

Freezing layers has been a widely used technique in fine-tuning and transfer learning approaches, as it enables models to retain pretrained knowledge while significantly reducing computational cost during optimization. A recent study [37] has demonstrated the effectiveness of this strategy in the context of score distillation. Pretrained diffusion models already train the convolutional layers to process corrupted images across various noise levels. Therefore, we assume that fully retraining these layers is often unnecessary and can lead to redundant computational cost.

Building on this observation, we apply a selective freezing strategy to our consistency model. Specifically, we freeze the convolutional layers of the backbone—which account for

85.8% of the total parameters—under the assumption that they have already learned rich image representations from prior diffusion model training. Since diffusion models are trained to predict local denoising directions along the trajectory, our consistency model can leverage this pretrained capability and instead focus on learning the global denoising scaling. Thus, only the remaining components are fine-tuned to align frozen features with the consistency objective, reducing the number of trainable parameters, lowering optimization burden, and enabling more efficient and rapid distillation.

This not only speeds up convergence but also mitigates the risk of overfitting when training on limited data or with constrained compute budgets. In practice, we find that this approach achieves comparable or better quality than full fine-tuning, while reducing both memory usage and wall-clock training time.

B. Stabilize Unbiased Score Estimation

In our proposed method, the reverse-time reference sample $\hat{\mathbf{x}}_r^\theta$ is gradually shifted toward the unbiased score estimate sample $\mathbf{x}_r$, which still suffers from high-variance estimation. To address the training variance in later stages of training, we employ a technique that utilizes reference batches for score estimation. Rather than relying on the single-sample estimate $-(\mathbf{x}_t - \mathbf{x}_0)/t^2$, we stabilize score estimation by approximating the conditional expectation $\mathbb{E}[-(\mathbf{x}_t - \mathbf{x}_0)/t^2 | \mathbf{x}_0]$ via Monte Carlo sampling over reference batches.

In contrast to the conventional approach, which estimates the score from a single sample and thus suffers from high variance, our reference-batch strategy leverages multiple samples within each minibatch to estimate the score expectation. By aggregating over diverse $\mathbf{x}_0$ instances, this method produces a more stable and statistically reliable score estimate. As a result, the model benefits from a smoother optimization trajectory and improved convergence stability, particularly in the later training stages when unbiased estimation becomes the dominant guidance signal

$$\nabla \log p(\mathbf{x}_t) = \mathbb{E}_{p(\mathbf{x}|\mathbf{x}_t)} [\nabla \log p(\mathbf{x}_t|\mathbf{x})] \quad (6)$$

$$= \mathbb{E}_{p(\mathbf{x})} \left[\frac{p(\mathbf{x}_t|\mathbf{x})}{p(\mathbf{x}_t)} \nabla \log p(\mathbf{x}_t|\mathbf{x}) \right] \quad (7)$$

$$\approx \frac{1}{N} \sum_{i=1}^N \frac{p(\mathbf{x}_t|\mathbf{x}^{(i)})}{p(\mathbf{x}_t)} \nabla \log p(\mathbf{x}_t|\mathbf{x}^{(i)}) \quad (8)$$

$$= \frac{1}{N} \sum_{i=1}^N \frac{p(\mathbf{x}_t|\mathbf{x}^{(i)})}{\frac{1}{N} \sum_{j=1}^N p(\mathbf{x}_t|\mathbf{x}^{(j)})} \nabla \log p(\mathbf{x}_t|\mathbf{x}^{(i)}) \quad (9)$$

$$= \frac{1}{N} \sum_{i=1}^N w^{(i)} \nabla \log p(\mathbf{x}_t|\mathbf{x}^{(i)}). \quad (10)$$

This technique was originally proposed in the context of diffusion models by Xu et al. [38], and later extended to consistency models by [31], which we follow in our framework. In practice, we implement this by treating each minibatch as a reference batch, allowing us to approximate the expectation without incurring additional computational cost. Specifically, we introduce the normalized importance weight

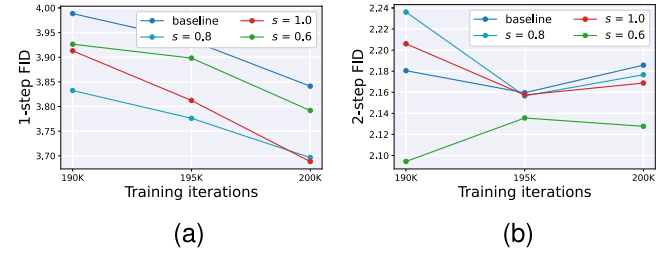


Fig. 5. FID scores (lower is better) under different scheduling configurations for interpolation growth. Our method generally achieves better or competitive performance compared to the baseline across varying hyperparameter s , demonstrating stable performance across the evaluated configurations. (a) One-step sampling results. (b) Two-step sampling results.

$w^{(i)} = p(\mathbf{x}_t|\mathbf{x}^{(i)}) / \frac{1}{N} \sum_{j=1}^N p(\mathbf{x}_t|\mathbf{x}^{(j)})$, enabling a weighted aggregation of gradients that emphasizes samples with higher likelihood. When using n reference samples, this Monte Carlo-based approach reduces the estimation variance proportionally to $1/n$, leading to a more reliable training signal.

This strategy is particularly effective in the later stages of training, where the variance of the unbiased score estimation tends to destabilize the training. In the early stage, the model is guided by student-generated samples, while in the later stage, it benefits from the stabilized unbiased estimator—ultimately leading to improved training stability and stronger final performance, as can be observed in the later phase of Fig. 5, where our method maintains a precise score trajectory compared to baselines.

VI. EXPERIMENTAL RESULTS

To evaluate the effectiveness of our proposed method, we conduct a series of comparative experiments using the recent efficient consistency model framework ECT [30] as a baseline. For a fair comparison, we integrate our method on top of the baseline without modifying its original configuration. These evaluations aim to highlight the performance and training efficiency improvements enabled by our framework.¹

A. Implementation Details

1) *Evaluation Protocol*: We evaluate sample quality using Fréchet inception distance (FID) [39], computed on 50k samples from both one-step and two-step sampling. All experiments follow standard evaluation protocols on CIFAR-10 [40] and ImageNet 64×64 [41], as in [24], [42], and [30].

2) *Training Details*: For CIFAR-10, we use the DDPM++ [4] architecture with pretrained EDM [42] weights, and for ImageNet 64×64 , we use EDM2-(S/M/L/XL) [43] with corresponding pretrained weights. All models are trained with a batch size of 128. Further details and extended results are provided in the Supplementary Material.

B. Discussion

1) *Variance-Reduced Consistency Learning*: To analyze the impact of vrCL in training stability, we compare our method

¹This article has supplementary downloadable material available at the project repository, provided by the authors. This includes the source code, setup scripts, and a README file. This material is 200 KB in size.

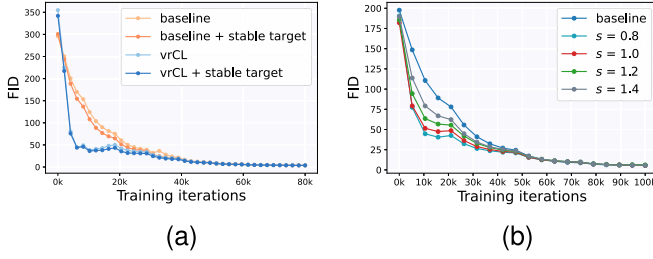


Fig. 6. Ablation study on training strategies for our model. (Left) Effect of vrCL during training. (Right) Effect of varying s on convergence speed and final FID. For a detailed comparison of final FID results, please refer to Fig. 5. (a) FID over iterations. (b) FID curves across s .



Fig. 7. Comparison of score estimation stability during training. The black curve represents the ground-truth score computed from a pretrained diffusion model, while the blue and yellow curves correspond to the unbiased estimator used in CT and our student-guided estimator, respectively. Shaded areas indicate the absolute estimation error relative to the ground truth. As shown, our method yields lower error than the CT estimator, particularly in early training. In later stages, the error remains suppressed due to the effect of the stabilized estimation technique, demonstrating improved consistency and reliability.

with the baseline. As illustrated in Fig. 6, vrCL achieves a markedly smoother optimization trajectory, reflected in faster and more consistent FID improvements throughout training. This stability is primarily attributed to our variance-reduced score estimation strategy, which mitigates the noisy gradients that often emerge in early training phases.

In addition, Fig. 6 demonstrates that such stability not only accelerates convergence but also consistently translates into superior final performance across a wide range of scheduling intensities, implying that the stable early-phase training provides the model more time to explore and converge toward an optimal solution.

In Fig. 7, we present a direct comparison of score estimation error between vrCL and CT (baseline), measured with respect to the teacher-guided estimation. While the error of CT remains persistently high, vrCL maintains significantly lower error throughout training, with the advantage being most pronounced in the early iterations.

We also investigate the effect of varying scheduling intensity, which controls the rate at which the interpolation coefficient β shifts from student-guided to unbiased samples. As shown in Fig. 6, higher scheduling intensity (rapid β growth) induces CT-like behavior by quickly prioritizing unbiased supervision, while lower intensity mimics CD-like behavior, retaining more low-variance, student-guided supervision for longer. This tunable balance enables vrCL to adapt between faster convergence and early-phase stability, effectively functioning as a continuous interpolation between the

TABLE II
ABLATION STUDIES OF FREEZING LAYERS ON CIFAR-10

Norm (7.9%)	Conv (85.8%)	QKV (2.1%)	Skip (4.0%)	FID ($\downarrow$)
✓	✓	✓	✓	3.63
✓	✗	✓	✓	3.57
✓	✗	✗	✓	3.66
✓	✗	✗	✗	3.65

two paradigms—CD and CT—thereby offering practitioners a principled way to tailor training dynamics to specific datasets and computational budgets.

2) *Stabilize Unbiased Score Estimation*: We evaluate the impact of reference-guided unbiased score estimation samples $\mathbf{x}_r$ by incorporating them into both the baseline model and our proposed framework. This technique leverages reference batches to reduce the variance of the unbiased score estimator $\mathbb{E}[-(\mathbf{x}_t - \mathbf{x}_0)/t^2]$ and is implemented using a minibatch as the reference, enabling a more accurate approximation of the expected score without relying on single-sample estimates. As shown in Fig. 7, this approach improves the stability of score estimation in the later stages of training, preventing the performance degradation that often arises from noisy supervision. Moreover, as demonstrated in Fig. 6, it results in more stable overall training dynamics and consistent improvements in FID performance. Importantly, the computation of the expected score using reference samples introduces only negligible additional overhead, as it reuses data already present in the minibatch, making the stability gains effectively cost-free in practical training scenarios.

3) *Partial Freezing*: Motivated by previous findings in score-based generative modeling, we adopt a partial freezing strategy where convolutional layers are fixed during training. We further evaluate the effects of freezing additional components, such as the qkv attention and skip-connection layers, to determine their individual contributions. Table II shows that, under our proposed vrCL framework, training only a small subset of the model can achieve better performance than full-layer training. This not only simplifies the optimization process but also enables efficient training with significantly fewer iterations. Since the pretrained convolutional layers have already captured rich image features, we retain them as-is and allow the remaining layers to adaptively realign these features to satisfy the consistency property of the model.

C. Comparison With Other Generative Methods

1) *Generative Performance*: To obtain optimal performance, we explore the effects of growth rate or β and timestep scheduling. As a result, we observe consistent improvements over the baseline, as demonstrated in Tables III and IV. As shown Table V, the training time remains consistent with that of baseline, while vrCL led to significant performance improvements.

Notably, we achieve a two-step FID of 2.02 on CIFAR-10 within just 100k iterations. In contrast, previous models required significantly more training steps and even larger architectures such as iCT-deep [25]. These results highlight the efficiency of our method in training consistency models while

TABLE III

QUANTITATIVE COMPARISON OF SAMPLE QUALITY ON CIFAR-10 DATASET, EVALUATED USING FID

Unconditional CIFAR-10			
METHOD	NFE ($\downarrow$)	FID ($\downarrow$)	
Diffusion models			
Score SDE [4]	2000	2.38	
Score SDE (deep) [4]	2000	2.20	
EDM [42]	35	2.04	
Flow Matching [44]	142	6.35	
PFGM [45]	110	2.35	
Joint Training			
Diffusion GAN [46]	4	3.75	
Diffusion StyleGAN [47]	1	3.19	
StyleGAN-XL [48]	1	1.52	
CTM [27]	1	1.87	
Diff-Instruct [18]	1	4.53	
DMD [22]	1	3.77	
SiD [23]	1	1.92	
Score Distillation			
DFNO (LPIPS) [49]	1	3.78	
2-Rectified Flow [50]	1	4.85	
PID (LPIPS) [51]	1	3.92	
PD [17]	1	8.34	
PD (2-step)	2	5.58	
TRACT [52]	1	3.78	
TRACT (2-step)	2	3.32	
CD (LPIPS) [24]	1	3.55	
CD (2-step)	2	2.93	
sCD [29]	1	3.66	
sCD (2-step)	2	2.52	
Consistency Training			
	Iter	NFE ($\downarrow$)	FID ($\downarrow$)
ACT [53]	-	1	6.4
ACT-Aug [53]	300k	1	6.0
CT [24]	800k	1	8.70
CT (2-step)	800k	2	5.83
iCT [25]	400k	1	2.83
iCT (2-step)	400k	2	2.46
iCT-deep [25]	400k	1	2.51
iCT-deep (2-step)	400k	2	2.24
ECT [30]	100k	1	4.54
ECT (2-step)	100k	2	2.20
vrCL (ours)	100k	1	3.16
vrCL (ours) (2-step)	100k	2	2.02

TABLE IV

QUANTITATIVE COMPARISON OF SAMPLE QUALITY ON IMAGENET 64×64 DATASET, EVALUATED USING FID

Class-Conditional ImageNet 64×64			
METHOD	NFE ($\downarrow$)	FID ($\downarrow$)	
Diffusion models			
ADM [3]	250	2.07	
RIN [54]	1000	1.23	
EDM (Heun) [42]	79	2.44	
EDM2 (Heun) [43]	63	1.33	
Joint Training			
StyleGAN-XL [48]	1	1.52	
Diff-Instruct [18]	1	5.57	
EMD [55]	1	2.20	
DMD [22]	1	2.62	
DMD2 [56]	1	1.28	
SiD [23]	1	1.52	
CTM [27]	1	1.92	
Moment Matching [57]	1	3.00	
Score Distillation			
DFNO (LPIPS) [49]	1	7.83	
PID (LPIPS) [51]	1	9.49	
PD [17]	1	15.39	
PD (2-step)	2	8.95	
TRACT [52]	1	7.43	
TRACT (2-step)	2	4.97	
CD (LPIPS) [24]	1	6.20	
CD (2-step)	2	4.70	
MultiStep-CD [58]	1	3.20	
MultiStep-CD (2-step)	2	1.90	
sCD [29]	1	2.97	
sCD (2-step)	2	2.07	
Consistency Training			
	Iter	NFE ($\downarrow$)	FID ($\downarrow$)
ACT [53]	400k	1	10.6
CT (LPIPS) [24]	800k	1	13.0
CT (2-step)	800k	2	11.1
iCT [25]	800k	1	4.02
iCT (2-step)	800k	2	3.20
iCT-deep [25]	800k	1	3.25
iCT-deep (2-step)	800k	2	2.77
ECT [30]	100k	1	5.51
ECT (2-step)	100k	2	3.18
vrCL (ours)	100k	1	5.17
vrCL (ours) (2-step)	100k	2	3.23

narrowing the gap with joint training approaches. Thanks to the fast convergence and efficient design.

2) *Comparison With the Baseline Across Training Budgets:* In addition, to evaluate the efficiency and scalability of our method, we compare its performance with the recent efficient consistency model framework ECT [30], which serves as our baseline, under fixed training budgets of 100, 200, and 400k iterations. In these experiments, the *relative speed of the β -shift is kept identical* by fixing the scaling parameters

s and n . The results demonstrate that vrCL consistently preserves its performance advantages without degradation even at larger training budgets. Furthermore, as β approaches 1 in the later stage of training, the use of self-referential samples is largely suppressed, indicating that the proposed framework effectively avoids bias accumulation and long-horizon model collapse. Importantly, while the baseline values are directly taken from the ECT paper, we re-tune the timestep scheduling ratio r/t and growth rate β for our method to optimize

TABLE V
FID (↓) SCORE FOR ONE-STEP AND TWO-STEP SAMPLING AT
DIFFERENT TRAINING ITERATIONS

Method		100k	200k	400k
1-step	ECT [30]	4.54	3.86	3.60
	Ours	3.16	3.06	2.96
2-step	ECT [30]	2.20	2.15	2.11
	Ours	2.02	2.01	2.01

TABLE VI
COMPARISON OF VRCL WITH BASELINE [30] ON IMAGENET 64×64
ACROSS DIFFERENT EDM2 MODEL SCALES

Method		EDM2-S	EDM2-M	EDM2-L	EDM2-XL
1-step	ECT [30]	5.51	3.67	3.55	3.35
	Ours	5.17	3.54	3.33	3.31
2-step	ECT [30]	3.18	2.35	2.14	1.96
	Ours	3.23	2.18	2.11	1.96

performance at each budget. As shown in Table V, our method consistently outperforms ECT, with particularly pronounced gains in the low-iteration regime.

In addition, we further validate the scalability of vrCL across model capacities by comparing it with the baseline on multiple EDM2 architectures (EDM2-S/M/L/XL), as summarized in Table VI. Across all model scales and both one-step and two-step sampling settings, vrCL consistently matches or outperforms the baseline, indicating that the proposed variance reduction strategy generalizes well to larger models and does not rely on a specific architectural configuration. These results further demonstrate that the effectiveness of vrCL is preserved not only across training budgets but also across model capacities, suggesting consistent performance under the evaluated settings and indicating its practical applicability.

VII. LIMITATIONS

Our training framework gradually transitions from a pretrained diffusion model to a consistency model via student-guided reverse-time references. Prolonged reliance on these student-generated references in later stages may introduce bias, as the model increasingly reflects global dynamics rather than local PF-ODE behavior. To alleviate this issue, we carefully control the growth rate of β . This added complexity is modest compared to the overall hyperparameter scale of modern generative models. In addition, we note that our experimental validation is primarily conducted at lower resolutions (e.g., 32×32 and 64×64), and evaluation at higher resolutions is not included in the current work. While we expect the proposed framework to generalize to higher resolutions, verifying this remains an important direction for future work.

VIII. CONCLUSION

We propose vrCL, a unified training framework for consistency models that eliminates the need for evaluating pretrained diffusion models during training while maintaining stability and efficiency. By integrating the strengths of CD

and CT, vrCL constructs reverse-time samples via student-guided parameterization and interpolates them with unbiased estimates, enabling low-variance supervision without additional architectural complexity. Our framework preserves the simplicity of one-step decoding, supports scalability across diverse generative settings, and reduces training cost through a selective freezing strategy. In addition, minibatch-based score estimation further stabilizes the supervision signal, making vrCL an efficient and practical approach for stable, teacher-free generative modeling.

REFERENCES

- [1] B. B. Moser, A. S. Shanbhag, F. Raue, S. Frolov, S. Palacio, and A. Dengel, "Diffusion models, image super-resolution, and everything: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 11793–11813, Jul. 2025.
- [2] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020.
- [3] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 8780–8794.
- [4] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," 2020, *arXiv:2011.13456*.
- [5] Z. Kong and W. Ping, "On fast sampling of diffusion probabilistic models," 2021, *arXiv:2106.00132*.
- [6] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, "DPM-solver: A fast ODE solver for diffusion probabilistic model sampling in around 10 steps," 2022, *arXiv:2206.00927*.
- [7] Q. Zhang and Y. Chen, "Fast sampling of diffusion models with exponential integrator," 2022, *arXiv:2204.13902*.
- [8] Z. Zhou, D. Chen, C. Wang, and C. Chen, "Fast ODE-based sampling for diffusion models in around 5 steps," 2023, *arXiv:2312.00094*.
- [9] S. Xue et al., "SA-Solver: Stochastic Adams solver for fast sampling of diffusion models," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2024.
- [10] H. Huang, L. Sun, B. Du, and W. Lv, "Learning joint 2-D and 3-D graph diffusion models for complete molecule generation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 11857–11871, Sep. 2024.
- [11] N. Huang et al., "DiffStyler: Controllable dual diffusion for text-driven image stylization," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3370–3383, Feb. 2025.
- [12] Z. Jiang et al., "RealignDiff: Boosting text-to-image diffusion model with coarse-to-fine semantic realignment," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 19010–19023, Oct. 2025.
- [13] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4713–4726, Apr. 2023.
- [14] A. Hatamizadeh, J. Song, G. Liu, J. Kautz, and A. Vahdat, "DiffT: Diffusion vision transformers for image generation," 2023, *arXiv:2312.02139*.
- [15] A. Ramesh et al., "Zero-shot text-to-image generation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8821–8831.
- [16] E. Luhman and T. Luhman, "Knowledge distillation in iterative generative models for improved sampling speed," 2021, *arXiv:2101.02388*.
- [17] T. Salimans and J. Ho, "Progressive distillation for fast sampling of diffusion models," 2022, *arXiv:2202.00512*.
- [18] W. Luo, T. Hu, S. Zhang, J. Sun, Z. Li, and Z. Zhang, "Diff-Instruct: A universal approach for transferring knowledge from pre-trained diffusion models," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2023.
- [19] J. Gu, S. Zhai, Y. Zhang, L. Liu, and J. Susskind, "BOOT: Data-free distillation of denoising diffusion models with bootstrapping," in *Proc. ICML Workshop Structured Probabilistic Inference Generative Model*, 2023.
- [20] A. Sauer, D. Lorenz, A. Blattmann, and R. Rombach, "Adversarial diffusion distillation," 2023, *arXiv:2311.17042*.
- [21] Z. Geng, J. Z. Kolter, and A. Pokle, "One-step diffusion distillation via deep equilibrium models," in *Proc. Adv. Neural Inf. Process. Syst.*, 36, 2023, pp. 41914–41931.
- [22] T. Yin et al., "One-step diffusion with distribution matching distillation," 2023, *arXiv:2311.18828*.
- [23] M. Zhou, H. Zheng, Z. Wang, M. Yin, and H. Huang, "Score identity distillation: Exponentially fast distillation of pretrained diffusion models for one-step generation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024, pp. 62307–62331.

- [24] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, "Consistency models," 2023, *arXiv:2303.01469*.
- [25] Y. Song and P. Dhariwal, "Improved techniques for training consistency models," 2023, *arXiv:2310.14189*.
- [26] S. Luo et al., "LCM-LoRA: A universal stable-diffusion acceleration module," 2023, *arXiv:2311.05556*.
- [27] D. Kim et al., "Consistency trajectory models: Learning probability flow ODE trajectory of diffusion," in *Proc. Int. Conf. Learn. Represent (ICLR)*, 2023.
- [28] F.-Y. Wang et al., "Phased consistency models," 2024, *arXiv:2405.18407*.
- [29] C. Lu and Y. Song, "Simplifying, stabilizing and scaling continuous-time consistency models," 2024, *arXiv:2410.11081*.
- [30] Z. Geng, A. Pople, W. Luo, J. Lin, and J. Z. Kolter, "Consistency models made easy," 2024, *arXiv:2406.14548*.
- [31] F.-Y. Wang, Z. Geng, and H. Li, "Stable consistency tuning: Understanding and improving consistency models," 2024, *arXiv:2410.18958*.
- [32] J. Sohl-Dickstein, "Deep unsupervised learning using nonequilibrium thermodynamics," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024, pp. 2256–2265.
- [33] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10674–10685.
- [34] C. Saharia et al., "Photorealistic text-to-image diffusion models with deep language understanding," in *Proc. NIPS*, vol. 35, 2022, pp. 36479–36494.
- [35] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with CLIP latents," 2022, *arXiv:2204.06125*.
- [36] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, "DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models," 2022, *arXiv:2211.01095*.
- [37] B. Zheng and T. Yang, "Diffusion models are innate one-step generators," 2024, *arXiv:2405.20750*.
- [38] Y. Xu, S. Tong, and T. Jaakkola, "Stable target field for reduced variance score estimation in diffusion models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023. [Online]. Available: <https://openreview.net/forum?id=WmlwYTd0YTF>
- [39] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 6626–6637.
- [40] A. Krizhevsky, (2009). *Learning Multiple Layers of Features From Tiny Images*. [Online]. Available: <https://www.cs.toronto.edu/>
- [41] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
- [42] T. Aila, M. Aittala, T. Karras, and S. Laine, "Elucidating the design space of diffusion-based generative models," in *Proc. Adv. Neural Inf. Process. Syst.* 35, 2022, pp. 26565–26577.
- [43] T. Karras, M. Aittala, J. Lehtinen, J. Hellsten, T. Aila, and S. Laine, "Analyzing and improving the training dynamics of diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 24174–24184.
- [44] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," 2022, *arXiv:2210.02747*.
- [45] T. Jaakkola, Z. Liu, M. Tegmark, and Y. Xu, "Poisson flow generative models," in *Proc. Adv. Neural Inf. Process. Syst.* 35, 2022, pp. 16782–16795.
- [46] Z. Xiao, K. Kreis, and A. Vahdat, "Tackling the generative learning trilemma with denoising diffusion GANs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021. [Online]. Available: <https://openreview.net/forum?id=JprM0p-q0Co>
- [47] Z. Wang, H. Zheng, P. He, W. Chen, and M. Zhou, "Diffusion-GAN: Training GANs with diffusion," 2022, *arXiv:2206.02262*.
- [48] A. Sauer, K. Schwarz, and A. Geiger, "StyleGAN-XL: Scaling stylegan to large diverse datasets," in *Proc. ACM SIGGRAPH 2022 Conf.*, 2022.
- [49] H. Zheng, W. Nie, A. Vahdat, K. Azizzadenesheli, and A. Anandkumar, "Fast sampling of diffusion models via operator learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022, pp. 42390–42402.
- [50] X. Liu, C. Gong, and Q. Liu, "Flow straight and fast: Learning to generate and transfer data with rectified flow," 2022, *arXiv:2209.03003*.
- [51] J. T. J. Tee, K. Zhang, H. S. Yoon, D. Gowda, C. Kim, and C. D. Yoo, "Physics informed distillation for diffusion models," *Trans. Mach. Learn. Res.*, Jun. 2024. [Online]. Available: <https://openreview.net/forum?id=rOvaUsF996>
- [52] D. Berthelot et al., "TRACT: Denoising diffusion models with transitive closure time-distillation," 2023, *arXiv:2303.04248*.
- [53] F. Kong et al., "ACT-diffusion: Efficient adversarial consistency training for one-step diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 8890–8899.
- [54] A. Jabri, D. Fleet, and T. Chen, "Scalable adaptive computation for iterative generation," 2022, *arXiv:2212.11972*.
- [55] S. Xie et al., "EM distillation for one-step diffusion models," 2024, *arXiv:2405.16852*.
- [56] T. Yin et al., "Improved distribution matching distillation for fast image synthesis," 2024, *arXiv:2405.14867*.
- [57] T. Salimans, T. Mensink, J. Heek, and E. Hoogetboom, "Multistep distillation of diffusion models via moment matching," 2024, *arXiv:2406.04103*.
- [58] J. Heek, E. Hoogetboom, and T. Salimans, "Multistep consistency models," 2024, *arXiv:2403.06807*.



Useok Choi received the B.S. degree in software convergence from Kyung Hee University, Republic of Korea, in 2026. He is currently pursuing the joint M.S. and Ph.D. degrees at the Department of Artificial Intelligence, Yonsei University, Republic of Korea.

His research interests include generative modeling, diffusion and flow models, knowledge distillation, and efficient deep learning.



Seungjoon Lee is currently pursuing the degree at the Departments of Software Convergence and Industrial Management Engineering, Kyung Hee University, Yongin, Republic of Korea.

His research interests include efficient generative models, knowledge distillation, explainable artificial intelligence, computer vision, and multimodal perception.



MyeongAh Cho received the B.S. degree in electronic engineering from Kyung Hee University, South Korea, in 2018, and the M.S. and Ph.D. degrees in electrical and electronic engineering from Yonsei University, South Korea, in 2023.

She is currently an Assistant Professor with the Department of Software Convergence, Kyung Hee University. Her recent work includes video anomaly detection, weather removal, and domain generalization. Her research interests lie in the areas of computer vision, computational photography, machine learning, and deep learning, with a particular focus on visual representation learning across diverse modalities, datasets, and levels of supervision.